

Post-Detection report for 2025 IAA SETI Committee meeting

1 Oct 2025, Sydney & online

Report given remotely by Kathryn Denning. (Associate Professor, Anthropology, York University, IAA SETI Committee member since 2005, SETI Institute Science Advisory Board member). kdenning@yorku.ca

Contents of today's update

1. Overview and Context
2. Post-detection-related Abstracts from IAC 2025
3. Some relevant recent, ongoing, and future activities of SETI Committee members, including new research groups, projects, talk series, conferences and symposia with post-detection relevance
4. NASA developments: Communicating Discoveries workshop report, NASA-DARES, etc.
5. Some recent and forthcoming publications of relevance
6. Revision of the “Declaration of Principles Concerning the Conduct of the Search for ExtraTerrestrial Intelligence (SETI)” and next steps: Specific points to ponder: PD Subcommittee reboot; Best Practices; etc.
7. Abstracts of Protocols Task Group’s papers in IAC Baku 2023, IAC Milan 2024 and IAC Sydney 2025.

1. Overview and Context

As in my last update, we do not currently have a formally organized Post-Detection Working Group or Subcommittee within the SETI Committee, but this could change in 2026 or 2027.

Meanwhile:

- a small task group within the Committee has been continuing the consultation and revision process for the “Declaration of Principles Concerning the Conduct of the Search for ExtraTerrestrial Intelligence (SETI)” (aka Protocols)
- some Committee members and other researchers from outside the Committee have been very active on post-detection-related topics

Recent post-detection updates to the Committee are available for reference: [2024](#), [2023](#), [2022](#) and [2021](#). These provide background and context for this year’s report.

We all know that 2025 has been a turbulent year for science and for international relations: much has changed since my last report. It is more important than ever to have longstanding international committees which work across borders and disciplines to promote the understanding and practice of science, the open sharing of scientific research, and multilateralism in decisions affecting our planet and the future of all humanity. And indeed, this standing SETI Committee of the International Academy of Astronautics has been endeavouring to do such work since it was founded in 1974 during the Cold War, starting with eight members from seven countries. The Committee was established following years of preliminary IAA discussions that began with a session in Madrid in October 1966.¹

This is a good moment to look back. When the Committee's first Chair, Rudolf Pešek (Czechoslovakia), reviewed ten years of IAA CETI² work from 1965-1976, he emphasized that the essential function of the Committee was to bridge between the CETI programs of different nations, noting that the work was "by its very nature, international in character both as regards its possible accomplishment and its implication for all mankind."³ This was his vision from the beginning: to unify conversations about CETI which were otherwise taking place only inside national borders. Pešek worked steadily on that goal in difficult times, even after the derailing of his four years of laborious planning for the first major international CETI symposium in Prague scheduled for Sept 1969: the symposium was cancelled due to the Aug 1968 invasion of Czechoslovakia by the Soviet Union and allies. But the planning and networking efforts were not in vain... the substantial Soviet-American Byurakan meeting of 1971 built on those foundations.⁴

Pešek carefully situated his reflection in the context of concurrent science, which included the first photographs of the surfaces of Mars and Venus, the Viking missions searching for life on Mars, the Venera missions, Pioneer 10 and 11 with their plaques, the 1974 Arecibo transmission, and prospective studies including the Cyclops report, and a 1975 study which predicted the first crewed interstellar starship would be launched in 2025. In addition to the story of the evolving science and technology and optimistic forecasts, Pešek also emphasized a constant – that starting with the earliest IAA CETI discussions in 1965, **the impact of CETI on humanity was recognized as an essential subject for discussion at international symposia, in tandem with the science.**

Pešek passed away in 1989, the year John Billingham – a founding IAA CETI Committee member and the second Chair – launched the first Protocols with Michael Michaud, Jill Tarter, and colleagues.⁵ Soon after, Billingham and colleagues were forging ahead with a key aspect of the Committee's vision: interdisciplinary, international workshops focused on SETI and Society. This, too, was derailed by politics for a time. Although three NASA-funded

¹ R. Pešek 1979 p 3. "Activities of the IAA CETI committee from 1965--1976 and CETI outlook" *Acta Astronautica* vol 6 pp 3-9. [Link](#). See also L. Walton and S. Dumas 2015 "The History of the SETI Committee: The Early Years", IAC-15,A4,2,1,x29264. [Link](#).

² Initially "Communicating" with Extraterrestrial Intelligence, later changed to SETI for "Search"

³ R. Pešek 1979 p 3.

⁴ G. Radulescu tells the story in "The Radio Universe, the Cold War, and Universal Communication" (2025) online at the Smithsonian National Air and Space Museum ([link](#)). Regarding the cancellation of the Prague symposium, Pešek's 1979 report makes only this restrained, very careful statement: "For different reasons, we have postponed the Symposium several times." This is an excellent illustration of why we need historians to tell us the whole tale of SETI. See also R. Charbonneau's recent book: *Mixed Signals: Alien Communication Across the Iron Curtain* ([interview](#) and [book](#)) and S. Dick's April 1993 paper as HRMS historian: "The Search for Extraterrestrial Intelligence and the NASA High Resolution Microwave Survey (HRMS): Historical Perspectives" in *Space Science Reviews* 64:93-139 ([Link](#)). In a similar vein, anyone who may be reading this report years in the future and seeking context may find it particularly useful to review the dramatic changes in 2025 to government-funded space science in the USA.

⁵ J.C. Tarter and M.A. Michaud eds, 1990 "SETI Post-Detection Protocol", special issue of *Acta Astronautica* 21(2). [Link](#).

workshops on the subject were held in California in 1991 and 1992 and a report produced – published years later by the SETI Institute⁶ -- subsequent plans were sidelined when NASA's funding for the HRMS observational SETI program was axed in October 1993.⁷ The newly established SETI Institute focused on keeping the observational SETI program alive as Project Phoenix, and overall, the interdisciplinary societal research didn't keep pace. Billingham and colleagues' long-established plan for an IAA SETI Committee-hosted 'International Conference on SETI and Society' to be held in Chamonix, France in June 1995 was initially postponed and eventually definitively cancelled due to lack of funding, despite attempts at private fundraising: by 1995, the only option left was for the Committee to encourage the original invitees to maintain their interest in this area and find a way to present their individual work at the IAC meetings instead.⁸ This was, inevitably, not as effective as the original plan would have been. Nonetheless, once again, the foundations carefully built by Pešek, Billingham, and colleagues meant that the ideals and scope of the vision remained. The SETI research community kept alive the principle that interdisciplinary work and international intentions both matter, and periodically convened smaller workshops on these themes.⁹ And now, in the 2020s, the next generation of researchers is taking up the challenge with new energy and vision.

Today, conversations and research on the social implications and cultural aspects of SETI and astrobiology happen in many places: the annual International Astronautical Congress is only one of them, and the work presented each year at the IAC represents only a portion of the relevant bodies of work. Partly this is because the high IAC registration fee is simply prohibitive for many researchers (despite the discounts for students and teachers), and the Congress is still a primarily in-person event requiring travel and in-person participation. (Thankfully, our own Committee's annual meetings are hybrid.) This is a significant structural issue in a world where international researchers can now easily meet instead frequently and for free on Zoom, or with lower registration costs, or in smaller regional meetings, to discuss their work and build joint efforts. It is also true that very large long-lived international organizations may not be the most administratively agile. Both longstanding tradition and innovation each have their places in a research ecosystem, and so these developments can and do coexist, but the shifting patterns of work and communication in the 2020s will inevitably mean that much relevant interdisciplinary research related to technosignatures takes place outside the IAA SETI Committee and IAC per se. So, we must continue to work to connect and bridge those conversations in a spirit of collaboration. This already requires concerted effort, patience, determination, and relationship-building, and that will be a constant in the years and decades ahead too – but we have ample precedent for that within the Committee's history, as the remarks above hopefully show. It can be done.

In 2025, while technosignatures searches continue to intensify and expand (as covered in other reports to the Committee), the societal and media impact of potential discoveries continues to be evident. So far this year, several putative discoveries of evidence of extraterrestrial life have provoked headlines and debates: the interstellar object 3I/ATLAS, the exoplanet K2-18b, and the Perseverance rover's Sapphire Canyon rock sample on Mars. As usual, there are enthusiastic proponents and more skeptical or cautious scientists responding to claims, with press releases, presentations, substantial media coverage and social media uptake, memes and distortions. Casual readers without relevant doctorates could be excused for not always being sure what exactly has been

⁶ J. Billingham et al 1999, *Social Implications of the Detection of an Extraterrestrial Civilization: A Report on the Workshops on the Cultural Aspects of SETI*. Report initially prepared 1994. SETI Press, SETI Institute, Mountain View California.

⁷ J. Tarter and P. Backus 1995, "Final Technical Report NCC 2-336" on NASA Technical Reports Server. [Link](#)

⁸ Billingham correspondence, 1995.

⁹ Some of these are listed in Denning et al 2019 "SETI and Post-Detection: Towards a New Research Roadmap". IAC-19-A4.2.4 x52210. [IAF](#). [Full text](#).

found, or why and how it matters. Intense media interest has always been a feature of SETI and astrobiology research, dating back to Drake's Ozma project in 1961¹⁰, but the landscape is increasingly complex.

This pattern can be expected to continue and escalate, and it underscores the importance of ongoing discussions about how these situations are best handled by the international community of scientists and science communicators. Work in this area is intensifying with respect to both biosignatures and technosignatures science and to their zones of overlap. Our Committee's current process of updating the *Declaration of Principles* is taking place within this dynamic context, and clearly, an approved updated draft of the *Declaration* will not be the end of our efforts. As the current draft specifies, we need additional guidelines regarding best practices for safety for scientists, and for science communication and risk communication in relation to SETI, and we need to reboot a subcommittee devoted to these and other challenges.

Below, I provide updates for some elements which have been mentioned in previous reports, including PSETI, the SETI Post-Detection Hub, NASA Astrobiology activities, the ongoing work of individuals.

Next year's report in 2026 will incorporate some new elements, including **an imminent new development within the SETI Institute**. The SETI Institute is pleased to confirm that starting in autumn 2025, there will be a new sustained research initiative within their Carl Sagan Center for Research, focused upon 'discovery and society', encompassing SETI, astrobiology, post-detection, science communication, and related topics. Lucian Walkowicz and Chelsea Haramia will be co-directors and I (Kathryn Denning) will be senior advisor. This initiative is the culmination of years of planning and builds on the Institute's strengths and the legacies of its founders and members, including several previous Chairs and Co-Chairs of the IAA SETI Committee. We look forward to sharing more details with the SETI community soon, and to building a new concentration of interdisciplinary, international, anticipatory, collaborative research on subjects of direct relevance to the Committee, and, of course, to cooperation with related international initiatives.

2. Post-detection Related Abstracts from IAC 2025

Some IAC 2025 SETI symposium contributions more specifically focused on topics related to public reaction and related policy etc. include the following. Some talks listed below were withdrawn/not presented, but the abstracts are still of interest. As usual, all the technical papers in the [SETI Science and Technology](#) session are also ultimately relevant to post-detection as well, given that they explore the forms of detection which are now possible, or may be possible in future. See <https://iafastro.directory/iac/browse/IAC-25/A4/> for the symposium listings and see the IAFastro Digital Library to search for full papers / proceedings: <https://dl.iafastro.directory/>

- C. Oliver and S. Shostak: The case of the EQ Peg hoax 25 years ago: Would a replay in today's media landscape have the same result? [Abstract](#) and [Paper](#)
- L. Walton: "The history of the IAA SETI Permanent Committee – 2000-2009" [Abstract](#) and [Paper](#)
- M. Garrett, K. Denning, L.I. Tenning, C. Oliver: "SETI Post-Detection Protocols: Progress Towards a New Version" [Abstract](#) and [Paper](#)

¹⁰ F. Drake 1979 "A reminiscence of Project Ozma" in *Cosmic Search* 1(1). [Link](#).

- D. De Paulis, M.L. Jensvold, N. Cabrol (withdrawn): “Patterns in Perception of a Simulated Message from Space” [Abstract](#)
 - A. Harrington: “The International Institute of Space Law SETI Working Group: Introduction, Overview, and Findings So Far” [Abstract](#) and [Paper](#)
 - S. Shagun (withdrawn): “The Societal and Legal Consequences of Detecting Extraterrestrial Intelligence: A Governance Gap” [Abstract](#)
 - K. Genevieve: “Refiguring Communication in SETI: Cybernetics, Analogy, and Improvisation in the Search for Extraterrestrial Life” [Abstract](#) and [Paper](#)
 - S. Alexander: “Space and Interstellar Travel and What We Want to be Remembered For – The Case for Indigenous Knowledges for Communication with Prospective ET’s and for Human spaceflight purpose and a Collective Earth Message” [Abstract](#) and [Paper](#)
 - P. Musso et al: “A first contribution to the history of Italian SETI” [Abstract](#) and [Paper](#)
 - A. D. Harvey and D. Wood: “New Cosmic Perspectives: Indigenous Knowledge and SETI” [Abstract](#) and [Paper](#)
 - C. Haramia: “Sustainability Puzzles and the Extraterrestrial Turn” [Abstract](#) and [Paper](#)
 - A. Mohite and S. Ankam: “Beyond Human Bias: Developing Non-Anthropocentric Methods for Detecting Extraterrestrial Intelligence” [Abstract](#)
 - N. Kerketta: “Differences in linguistic cultural and moral values of a similarly intelligent extraterrestrial life-form” [Abstract](#)
 - D. De Paulis and A. Melis (withdrawn): “A Single Sky: a new simulated First-Contact scenario engaging the global scientific SETI network, and integrating deep space technologies into SETI search.” [Abstract](#)
 - K. Bayramova: “The role of consciousness in defining reality: a philosophical and scientific inquiry” [Abstract](#)
 - E. Norton-Fox: “Earth’s first encounter. Will making contact strengthen humanity or cause more harm than good, and if so, who makes the first move?” [Abstract](#)
 - R. Alazzam et al: “ The Search for Meaning: The Social Impact of SETI and Its Influence on Humanity” [Abstract](#)
 - P. Musso, N. Antonietti et al: “Five years of SETI at the InCosmiCon Research Center” [Abstract](#)
 - S. Raviraja: “Artificial Human Evolved Intelligence vs. Naturally Evolved Biological Intelligence: Interaction between the two and Game Theory” [Abstract](#)
-

3. Some relevant recent, ongoing, and upcoming post-detection-related activities of SETI Permanent Committee members and others (n.b. This list is far from complete.)

- The [SETI Institute](#) and [Breakthrough Listen](#) continue their extensive public outreach efforts.
- **Breakthrough Discuss 2025, Oxford and online:** “[Life As We Don't Yet Know It](#)” included sessions on “Forms of Non-Terrestrial Life”, “The Nature of Consciousness and Intelligence”, and “Detecting Life as We Do Not Know It” and was co-hosted by Dept of Physics, U. Oxford, Dept Physics and Astronomy at U. Manchester, the Cavendish Lab at U. Cambridge, and the SKA Observatory. Previous Discuss conferences back to 2016 are [here](#). As usual, the range of actual and potential types of discovery, and new technologies and forms of experimentation, all pose myriad questions for post-detection / discovery & society, although the latter are not usually explored in depth in this conference. [Talk Playlist](#).
- **PSETI (Penn State University SETI Center).** The [PSETI seminar series](#) run by Committee member [Jason Wright](#) and colleagues includes SETI & Society / post-detection topics – recordings are available. PSETI also sponsors the [Order of the Octopus](#), a large, thriving international community of early career researchers in SETI working across disciplines. PSETI projects have included a new Zotero bibliography including interdisciplinary SETI, building on the ADS [bibgroup: SETI](#) project started a few years ago¹¹ – contact J. Wright for details. Also of particular note is Wright’s forthcoming book, *SETI: Theory and Practice* (IOP Publishing / AAS e-Books) which is a broad overview of SETI, including its history, theory, practice, and social aspects, intended for advanced undergraduates in astronomy and graduate students in the astrobiological sciences.
- The third [PSETI Symposium 2025](#) was held 18 - 21 Aug 2025, online + in person in Pennsylvania. As always, it took a broad look at SETI, reporting on new search methods and instrumentation, theory, search results, and social aspects of the field from history to post-detection. Previous symposia have had a significant focus on post-detection and related topics in talks and breakout sessions, with multiple Committee members contributing, and 2025 continued that trend. The [website](#) includes [abstracts](#) and will eventually also include recordings of plenary talks. [Plenary talks](#) covered post-detection protocols updates, science communication and risk communication for astrobiology and SETI, METI and ethics, continuations of the Sign in Space project, sustainability, concepts of intelligence, and more, while [posters](#) also covered relevant terrain. Several post-detection-related breakout sessions were proposed -- and indeed, required coordination so that we would not compete against each other. [Full session descriptions are here](#). *SETI in the Real World: Post-Detection and More* (with A. Berea, J. Bimm, R. Charbonneau, K. Denning, C. Haramia, B.S. McConnell, G. Profitiliotis, A. Schwarz, L. Walkowicz, and L.I. Tennen) – a follow-up to previous sessions in [2022](#) and [2023](#) – focused on science and risk communication in an interactive exercise playing through the hypothetical example of a technological object in our solar system, briefly addressing space law, current social media, and the possibility space for future best practices for technosignatures discoveries. Related breakout sessions included *What Is It Like to Be an Extraterrestrial?* by D. De Paulis, C. Vidal and colleagues (an exercise focused on message construction and decipherment) and *All Worlds, All Times*, a live action role play exercise led by K. Genevieve, T. Fisher, G. Profitiliotis, A. Kovacevic, and colleagues.

¹¹ A. Reyes and J.T. Wright 2019 “Towards a Comprehensive Bibliography for SETI” <https://arxiv.org/abs/1908.02587>

- **The Technosignatures Seminar Series** at NASA Goddard run by Committee members Ravi Kopparapu and Jacob Haqq-Misra, continues its excellent work in featuring speakers on a variety of technosignatures topics, including SETI Committee members, and some talks directly relevant to post-detection. The website has a past and future speaker list and recordings of presentations back to 2020, which by now comprises a significant archive of presentations and discussions. To join the mailing list and receive reminders and Zoom access information for upcoming seminars, contact Committee member Ravi Kopparapu: ravikumar.kopparapu@nasa.gov
- **Daniela de Paulis and team, A Sign in Space** This award-winning interdisciplinary art project, presented to the Committee by our member de Paulis in 2021 and 2022 as work in progress, proceeded in 2023-2024 with the involvement of the SETI Institute and several observatories, and continues to be a great success, with thousands of citizen participants engaged. Multiple members of the SETI Committee and many other researchers were involved in different capacities. The work continues: recent talks by De Paulis may be found [here](#) and [here](#), and additional publications will follow.
- **The SETI Post-Detection Hub at the University of St. Andrews** Coordinator: IAA SETI Committee long-time member John Elliott (john.elliott@st-andrews.ac.uk). As previous reports detailed in 2023 and 2024, the SETI Post-Detection Hub at St. Andrews was launched in Nov 2022 with a focus on the potential future discovery of intelligent life beyond Earth, and the interdisciplinary work necessary to prepare responsibly for such an eventuality. The Hub continues to grow, and now has over 50 researchers from many disciplines, about half from the UK and half international, some with long histories of working on SETI topics, and others newer to the field. Many IAA SETI Committee members are now also [Hub members](#). Some Hub members are actively engaged in international conferences (e.g. PSETI, IAC, IAU404, etc.) and in Working Groups within the Hub, as well as in their own individual research and publication projects. The Hub now has a monthly internal newsletter, The Waterhole, edited by C. Oliver.

The Hub's activities in 2025 have included: a Post-Detection Futures workshop held in July 2025 at St. Andrews; engagement in media outreach and in science festivals; submissions to the UK House of Lords Select Committee on Engagement with Outer Space; ongoing engagement with the UK government's Space Directorate; a multi-authored white paper submitted to the NASA-DARES RFI; ongoing efforts towards funding; projects with a student group (The Order of the Crow); and the ongoing creation of The Post-Detection Toolkit, which combines potential future scenarios for our own world with SETI detection situations, to help the research community imagine post-detection challenges and responses (stay tuned for more). Please see the PDF from J. Elliott for additional details of Hub activities over the past year.

- **IISL Working Group on SETI & Law.** In Jan 2024, the International Institute of Space Law established a Working Group on SETI & Law under convenor Michael Bohlander. As of Sept 2024, it has gone forward under the leadership of Andrea Harrington and is scheduled to run until 2026. Details of the WG's focus and intended outputs are evolving slowly, but generally, the WG is focused upon legal issues that would follow a confirmed detection of extraterrestrial intelligence. Several IAA SETI Committee members are involved in the Working Group, including M. Garrett, L.I. Tennen, J. Elliott, J. Wright, and K. Denning. See Harrington's [IAC 2025 paper](#) for further information.

- The [InCosmiCon Research Center \(Intelligence in the Cosmic Context\)](#) was established in 2020 by Committee member Paolo Musso at the University of Insubria, and has since grown to over 50 members from 9 countries, and several Committee members, including Nicolò Antonietti, Yvan Dutil, Teófilo Vargas, Andrea Melis, and Stelio Montebugnoli. (The late Claudio Maccone was also very involved.) Its research areas include several aspects of SETI, as Musso and colleagues explains in their [2025 IAC paper](#).
- [ASPIRE One](#): Our member Anamaria Berea has been leading a project of the [Interstellar Foundation](#), working on “planetary messages in a bottle”. The [Aspire One](#) archive / record of carefully curated content was designed for future humans and encoded physically in an artifact designed to last for millennia, and delivered to the Moon in March 2024 by the Firefly Blue Ghost mission 1. [The team’s](#) larger project of building a generalized framework for planetary communication is described in a 2025 publication.¹² See Berea’s Sept 2025 [talk](#) for [The Technosignatures Seminar Series](#).
- [The Cosmic Footprint Society](#), launching October 2025, involves multiple Committee members and SETI community members, who co-signed or endorsed the open letter “A call to address humanity’s cosmic footprint” in July 2025.¹³ This initiative seeks to engage the UN in a comprehensive discussion about humanity’s intergenerational responsibilities and impacts beyond Earth: “we, stakeholders and specialists from various nations, involved in a broad spectrum of space-related disciplines and activities, collectively call to humankind, the nations of Earth, scholars, experts, policymakers, stakeholders, space-related corporations and ethics committees, to unite in recognizing the need to assess and responsibly address humanity’s impact beyond Earth.” The initiative advocates for “enduring strategies and structures designed to keep humanity accountable for its cosmic footprint over the long term” and the “establishment of an international registry of our anthropogenic footprint beyond Earth” in all domains, including orbital debris, spread of terrestrial lifeforms, and changes in Earth’s electromagnetic signature which could increase the likelihood of our detection by ETI. It will be interesting to see this initiative’s next steps. Those interested can read [the call](#), and [fill in a form](#) to endorse the initiative, join the society, and otherwise signal their interest.

Upcoming:

- [AbSciCon 2026](#) (17-22 May 2026, Madison Wisconsin) can be expected to include sessions on technosignatures, exoplanets, discovery communications, and the societal impacts of research that seeks and may find extraterrestrial life, as well as the presentation of NASA-DARES ([Decadal Astrobiology Research and Exploration Strategy](#)) which is currently in development in 2025. The conference this year will be somewhat hybrid, with in-person sessions, posters, etc and also online discussion sessions. Session proposals are due 8 Oct 2025 and paper abstracts are due in January 2026. In recent years, AbSciCon has had sessions at the intersection of extraterrestrial life, social impact, social science and humanities (including some organized by Committee members A. Berea and K. Denning) and on technosignatures (Committee member J. Haqq-Misra and colleagues) and similar sessions are anticipated this year.

¹² A. Berea, K. S. Lewis, B. Beinhoff, A. Kershenbaum, E. Sengsavang, N. Searra 2025, “Towards a Generalized Framework for Planetary Communication” in *Space Policy* 26 July 2025 <http://dx.doi.org/10.2139/ssrn.5002321> or [Preprint](#).

¹³ A. Normier, A. Boujibar, D. Boulesteix *et al.* 2025. “A call to address humanity’s cosmic footprint.” *Nature Astronomy* 9, 934–935. <https://doi.org/10.1038/s41550-025-02606-7> . Or [Full text preprint with endorsers](#).

- [IAUS 404: Advancing the Search for Technosignatures](#), online symposium, 2-6 March 2026: the SOC includes J. Haqq-Misra, R. Kopparapu, B. Villaroel, M. Dominik, G. Profitiliotis and S. Berdyugina, most of whom are SETI Committee members. The online-only mode is promising for full accessibility and international participation. The symposium's emphasis is primarily on ways of advancing the search for technosignatures, but some no doubt some interdisciplinary content will be included too. Abstracts are due 17 Oct 2025.
-

4. NASA Developments

2025 has, to put it mildly, been an eventful year for NASA and government-funded science in the USA, with actual and projected funding cuts, workforce reductions, program cancellations, and specific targeting / elimination of diversity and equity initiatives and climate science programs. Agency priorities are being significantly restructured under the new federal government since January 2025.

Last year, I reported that technosignature research was slowly being reintegrated into the NASA Astrobiology portfolio after a long hiatus, and that the Program leadership was focusing quite significantly on the challenges of communicating potential future discoveries of life beyond Earth to wide audiences, e.g. in workshops and in its Oct 2024 [Request for Information](#) for the upcoming NASA Decadal Astrobiology Research and Exploration Strategy. Today, the overall future of astrobiology science within NASA is unquestionably precarious, as the present White House proposed budget includes profound cuts within NASA, some of which have already started to be enacted, although they have not been passed by Congress. There have been significant losses of senior NASA personnel. Nonetheless, there are some developments to report which are positively relevant to the present subject of post-detection:

- The group-authored report for the large 2024 NASA Astrobiology workshop, *Communicating Discoveries in the Search for Life in the Universe* ([page](#)), will be published soon in *Astrobiology* (Nov 2025) and on ArXiv¹⁴. I briefly described the workshop in my [2024 report](#) for this Committee, and a talk was presented by Jordan Bimm (co-organizer and lead author of the report) at [PSETI 2025](#). The CDSLU workshop and report involved multiple technosignatures researchers and members of this Committee, and highly relevant themes are covered in the report, including: the importance of effective collaboration between disciplines; full analysis of prior case studies of putative discoveries of extraterrestrial life; the support of targeted social research in science communication for astrobiology which addresses problems such as ambiguity, rapidly evolving science, misinformation, and perceived risk; ways for funding agencies to support verification and communication efforts; and, crucially, safety structures which protect scientists in the event of media frenzies.
- The [NASA Decadal Astrobiology Research and Exploration Strategy](#) process is now underway. Notably, the [2024 RFI](#) included the topic "Prepare for the Discovery of Life Beyond Earth and Subsequent Post-Discovery Activities" and accordingly, several White Papers along those lines were submitted in early 2025, and summarized in the May 2025 RFI Findings Workshop by DARES Task Force 1 member Jordan Bimm under the heading "Astrobiology and Society" – see the recording [here](#) at time 2:30:10. (See also the recorded presentations from Sofia Sheikh and Anamaria Berea.) The next stage of NASA DARES is

¹⁴ J. Bimm, M. Voytek, C. Scharf et al. forthcoming 2025. "Communicating Discoveries in the Search for Life in the Universe (CDSLU) Workshop Report", to appear in *Astrobiology* Volume 25, No. 11, Nov 2025, and on ArXiv.

about to begin, with the selection of Task Force 2 members. Community workshops are anticipated throughout fall 2025 – dates are not yet certain, but updates will appear on the [DARES website](#) – and a draft will be presented to the community for comment in early 2026, with the final Strategy scheduled for presentation at AbSciCon in 2026. Ultimately, the DARES strategic recommendations can reasonably be anticipated to include the development of a roadmap for cumulative, sustained research about post-discovery considerations such as optimizing public communication... but we will see next year. The NASA Astrobiology Community Update of [August 2025](#) affirmed the program’s ongoing commitment to the matter of ‘discovery communications’ and to interdisciplinary work in this area.

5. Some Recent and Forthcoming Publications of Relevance (not comprehensive!)

- D. Albergaria, P. Russo, I. Smeets, T. Heenatigala, D. Vetter 2025. “Communicating astrobiology and the search for life elsewhere: Speculations and promises of a developing scientific field in newspapers, press releases and papers.” *PLoS One* 20(7): e0328766. <https://doi.org/10.1371/journal.pone.0328766>
- D. Albergaria, P. Russo, T. Heenatigala, K. Fujishima, D. Scalice 2025. “Executive Summary of *We Found Extraterrestrial Life!* A Lorentz Center Workshop about Communicating Astrobiology with the Public.” Workshop held Sept 2024. [Link](#). See also associated brief [report](#) and [website](#).
- A. Berea et al 2025. “Towards a generalized framework for planetary communication” in *Space Policy* online 26 July 2025, <https://doi.org/10.1016/j.spacepol.2025.101711>
- J. Bimm, M. Voytek, C. Scharf et al. forthcoming 2025. “Communicating Discoveries in the Search for Life in the Universe (CDSLU) Workshop Report”, to appear in *Astrobiology* Volume 25, No. 11, Nov 2025, and on ArXiv
- R. Charbonneau 2024. *Mixed Signals: Alien Communication Across the Iron Curtain*. Polity Press. ([interview](#) and [book](#)).
- R. Charbonneau 2024. “Why did Earth’s first radio message to alien civilizations leave out half of humanity?” in *Nature* 12 Nov 2024. www.nature.com/articles/d41586-024-03677-8
- O. Eldadi, G. Tenenbaum, A. Loeb 2025. “The Adaptive Communication Framework (ACF) for Extraterrestrial Intelligence Discovery.” Preprint. [10.48550/arXiv.2508.18287](https://arxiv.org/abs/10.48550/arXiv.2508.18287)
- A. Fraknoi 2025. “A Selection of Science Fiction Stories with Reasonable Science about SETI, Techno-signatures, & the Fermi Paradox”, updated in 2025. Available at <https://bit.ly/sfaboutseti>
- K. Genevieve et al 2025. “SETI Post-Detection Futures: Directions for Technosignature Research and Readiness” (White paper for NASA DARES RFI 2025) <https://arxiv.org/abs/2507.11587v1>
- C. Haramia, E. A. LaFleche, J. De Marines, M. L. Wong 2025. “How to chart a moral future for space exploration” in *Nature* 8 July 2025. www.nature.com/articles/d41586-025-02070-3
- J. Haqq-Misra, G. Profitiliotis, R. Kopparapu 2025. “Projections of Earth’s technosphere: Scenario modeling, worldbuilding, and overview of remotely detectable technosignatures” in *Technological Forecasting and Social Change* Vol 218 <https://doi.org/10.1016/j.techfore.2025.124194>

- H. Landt, M. Dominik, C. Oliver, eds., forthcoming 2025. *(Toward) Discovery of Life beyond Earth and its Impact (IAU S387)*. Cambridge University Press. [Link](#). Proceedings of the [IAUS387 symposium](#), Durham April 2024.
- A. Normier *et al.* 2025 “A call to address humanity’s cosmic footprint” in *Nature Astronomy* 9: 934–935. <https://doi.org/10.1038/s41550-025-02606-7> . [Full text with contextual note](#).
- G. Profitiliotis 2024. "Moral Vistas to Xenic Beyonds: Fostering Moral Imagination to Pre-empt Monsterization in Future Encounters With Extraterrestrial Life", in *The Ethics Gap in the Engineering of the Future: Moral Challenges for the Technology of Tomorrow*, eds Spyridon Stelios, Kostas Theologou. Emerald Publishing. <https://doi.org/10.1108/978-1-83797-635-520241010>
- E. Puranen, J. Donovan, M. Ajevski 2025 “Who speaks for extraterrestrial ecosystems?: Why ET should have standing” in *Space Policy*, online 19 Sept 2025. <https://doi.org/10.1016/j.spacepol.2025.101722>
- G. Radulescu 2025, “The Radio Universe, the Cold War, and Universal Communication”, Smithsonian National Air and Space Museum 23 May 2025 ([link](#)).
- G. Radulescu 2024, *A history of communication with extraterrestrial intelligence (CETI): astronautics and radio astronomy across the Iron Curtain (1956–1976)*, PhD dissertation, Technical University of Berlin. [Link](#).
- G. Radulescu 2025, “From distance(s) to civilization(s): (Extra)terrestrial intelligence(s) of (post-) Soviet Armenian astronomy” in *Social Studies of Science*, online 18 March 2025 <https://doi.org/10.1177/03063127251324659>
- UNESCO 2024 (J. Gerards and N. Schattevoet), “The safety of scientific researchers: data, trends and a typology of threats” <https://unesdoc.unesco.org/ark:/48223/pf0000389907> . See also [UNESCO Programme on the Freedom and Safety of Scientists](#).
- D. Vakoch 2024, “A message of peace to Jupiter’s moon from actually all of humankind” in *Nature* 17 Dec 2024 www.nature.com/articles/d41586-024-04192-6 or [full text](#).
- J.T. Wright, forthcoming 2025. *SETI: Theory and Practice*. IOP Publishing / AAS e-Books.
-

6. The Protocols: Ongoing revision process for the “Declaration of Principles Concerning the Conduct of the Search for ExtraTerrestrial Intelligence (SETI)”

As Committee members know, a small Task Group (Michael Garrett, Carol Oliver, Les Tennen, Kathryn Denning, with input from additional Exec members Lori Walton and Andrew Siemion) has been working on a revision of the *Declaration of Principles* for several years, building on prior work from SETI Committee members.

Abstracts from our 2023, 2024, and 2025 IAC papers about this process are included below, with links to full-text versions for reference. Additional materials from Committee updates are here: [2023](#) and [2024](#).

My [2024 report](#) briefly covered the ‘History of Prior Work on SETI Post-Detection Protocols’, which may be useful to review. I also underscored the need for better archives of previous work. (We anticipate tackling this

challenge at the SETI Institute, appreciate that others have also started in on this work, and look forward to making progress with it.)

In 2025, members of the Committee and of the larger SETI Community received a request for detailed feedback on drafts in a structured process. Committee members have now received a copy of the current [25 Sept 2025](#) “DRAFT REVISED Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence (SETI) –2026 Update”. Other reports in today’s meeting will cover the details of the feedback process, the edits we made, and the next steps towards eventual approval and adoption of this new Declaration, and so I will not recap those here. Instead, I have three general points here, speaking as an individual:

- First, I am grateful to all those who took the time to provide their comments in the feedback process, and prior to that. **Thank you.** It is not possible to address all comments / include all remarks in a document of this nature, and even within the Task Group our opinions diverged at times, but I hope we collectively did a decent job of addressing the highest shared priorities of this research community today. I will keep in mind the comments made by Committee and community members and will continue to consider how to most constructively address longstanding and emerging debates in my own future work.
- Second, we have more work to do. As the current Draft indicates, **we need additional Best Practices guidelines to accompany the new Declaration.** The idea here (first floated by a team of Committee authors in 2019¹⁵) is that these could be less formal, readily produced and updated by small working groups. Best Practices for safety for scientists, for projects with students, for projects with citizen scientists, for data handling, and for communicating about technosignatures to broad audiences, may all be useful. The issue of safety for scientists is particularly significant and urgent. The forthcoming NASA Communicating Discoveries workshop report highlights this issue, and so does a significant UNESCO program and associated report.¹⁶ We should also begin work on best practices for potential discoveries of technosignatures within our own solar system in the form of physical objects, given the moving detection thresholds and potential forms of discovery which are now possible.¹⁷
- Third, **on the Post-Detection Sub-Committee and overall Committee composition:** The current [Draft Declaration](#) commits us to re-establishing a PD sub-committee after a long hiatus. This has seemed to be an uncontroversial proposition in the feedback exercises, i.e. it is well-supported. Previously, PD Sub-Committees existed within the Committee from about 1996 to 2014, with varying levels of activity and remits.
 - As I mentioned in my [2024 report](#) and previously, we need now to consider overall Committee composition. The Committee has many technical experts and astronomers, but – in contrast to the balance that used to exist years ago -- very few legal experts, professional science communicators, or science communication researchers, and comparatively few members in

¹⁵ Denning et al 2019 “SETI and Post-Detection: Towards a New Research Roadmap”. IAC-19-A4.2.4 x52210. [IAF. Full text.](#)

¹⁶ UNESCO 2024 (J. Gerards and N. Schattevoet), “The safety of scientific researchers: data, trends and a typology of threats” <https://unesdoc.unesco.org/ark:/48223/pf0000389907> . See also [UNESCO Programme on the Freedom and Safety of Scientists.](#)

¹⁷ J.R. Davenport, S.Z. Sheikh, S. Croft *et al* 2025. “Technosignature Searches of Interstellar Objects” <https://arxiv.org/abs/2508.16825>

humanities/social science. We do not, in short, presently have a critical mass of “SETI and Society” experts within the Committee. We therefore need an intentional approach to membership to build capacity. We also need an intentional approach to make the Committee more representative of the SETI research community as it is today: there is a *significant* gender skew in the Committee, and an even worse age/career stage skew. This should be rectified.

- To once again foreground an issue which has been mentioned often in the past within this Committee and which I covered in some detail last year: we need to carefully consider the limitations and actual capacities of a Post-detection Subcommittee with no funding of its own.

7. Abstracts of IAC papers by the Task Group, re: the Protocols revision

To be presented in the IAC Sydney [SETI & Society](#) session on 1 Oct 2025. On [IAF Digital Library server](#).

IAC-25-A4.2.4 (99704)

SETI Post-Detection Protocols: Progress Towards a New Version

Michael A. Garrett^{*a}, Kathryn Denning^b, Leslie I. Tennen^c, Carol Oliver^d

^a *University of Manchester, Jodrell Bank Centre for Astrophysics, UK, and University of Leiden, Leiden Observatory, NL. michael.garrett@manchester.ac.uk*

^b *Dept of Anthropology, York University, Toronto, Canada. kdenning@yorku.ca*

^c *Law Offices of Sterns and Tennen, Phoenix, Arizona, USA. ltennen@astrolaw.com*

^d *Australian Centre for Astrobiology, School of Biological, Earth, and Environmental Sciences, University of New South Wales, Australia. carol.oliver@unsw.edu.au*

** Corresponding Author*

Abstract

The International Academy of Astronautics (IAA) SETI Committee has long provided guiding principles for responding to a potential detection of a SETI signal. The foundational *Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence*, first formulated in 1989, has been widely recognised by the international scientific community. A supplemental set of draft protocols addressing the possibility of a reply to an extraterrestrial signal was prepared in 1995 by the IAA SETI Permanent Committee, with both documents presented in a position paper to the UN Committee on the Peaceful Uses of Outer Space in 2000.

In keeping with the evolving landscape of SETI research, the IAA *Declaration of Principles* was streamlined and updated in 2010. Recognising the need for continued adaptation, the IAA SETI Committee established a Task Group in 2022 to re-examine the protocols in light of recent advances in search methodologies, the expansion of international participation in SETI, and the increasing complexity of the global information environment. The Group recognises the living document nature of the protocols, which will require ongoing refinement to remain relevant and effective in a rapidly changing world.

A preliminary report was presented at the 2023 International Astronautical Congress (IAC) in Baku, outlining proposed revisions. A draft revised *Declaration of Principles* was presented at the IAC 2024 in Milan, and initial feedback was received from the community, particularly members of the IAA SETI Committee. Since then, we have continued to seek broader community input in a structured process, refining the proposed updates based on further discussions and consultations.

At the IAC 2025 in Sydney, we intend to present the outcomes of this process, including a new draft of the *Revised Declaration of Principles*, which will also be submitted to the IAA SETI Committee at its annual meeting during the Congress.

IAC-24-A4.2.6 x84417

The Future of the SETI Post-Detection Protocols: Progress Towards Revisions

Leslie I. Tennen^{a*}, Kathryn Denning^b, Carol Oliver^c, Michael A. Garrett^d

^a *Law Offices of Sterns and Tennen, Phoenix, Arizona, USA. ltennen@astrolaw.com*

^b *Dept of Anthropology, York University, Toronto, Canada. kdenning@yorku.ca*

^c *Australian Centre for Astrobiology, School of Biological, Earth, and Environmental Sciences, University of New South Wales, Australia. carol.oliver@unsw.edu.au*

^d *University of Manchester, Jodrell Bank Centre for Astrophysics, UK, and University of Leiden, Leiden Observatory, NL. michael.garrett@manchester.ac.uk*

** Corresponding Author*

Abstract

The International Academy of Astronautics has been the leading source of guiding principles to be followed in the event of a putative SETI signal detection. After many years of work, in 1989, the IAA formulated a set of Protocols, the “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”, which received widespread acceptance by the international scientific community. The IAA prepared a supplemental set of draft protocols in 1995 relating to the possible reply to a confirmed communication from an ETI, and both documents were presented in a position paper to the UN Committee on the Peaceful Uses of Outer Space.

The Search for Extra-Terrestrial Intelligence takes place in a dynamic context, and the Protocols are always a “work in progress” that must adapt to and reflect constantly changing scientific and social environments. Thus, the IAA SETI Protocols were streamlined and updated in 2010 after a multi-year process. Subsequently, the IAA SETI Committee has sought to review and revise the Protocols to keep pace with the expansion of search methods, the increased number of facilities and personnel conducting search activities worldwide, and the rapidly evolving information environment. In 2022, a Task Group was formed for this purpose, and a preliminary report was presented the following year during the 74th International Astronautical Congress in Baku. The relevance of this endeavour has been underscored in the past year by the deepening interest in what happens after the discovery of life beyond Earth, as evidenced by the 2024 workshops and symposia sponsored by the NASA Astrobiology Program, and the IAU/Kavli Foundation, the work of the SETI Post-Detection Hub at the University of St. Andrews, and the working group on “SETI and the Law” newly established by the International Institute of Space Law.

This report updates the activities of the Task Group following the 2023 Baku IAC, including further community consultations and proposed text, as we work towards a combination of timeless Principles reflecting the original intentions of the 1989 Protocols and supplementary Codes of Conduct and Best Practices documents that can provide additional guidelines that can be updated regularly to respond to changing circumstances.

[IAC-23-A4.2.3 x79618](#)

Revising the SETI Post-Detection Protocols for the 2020s and Beyond: A Report on Work in Progress.

Carol Oliver, Kathryn Denning, Leslie I. Tennen, Michael A. Garrett

Australian Centre for Astrobiology, School of Biological, Earth, and Environmental Sciences, University of New South Wales, Australia, carol.oliver@unsw.edu.au

Dept of Anthropology, York University, Toronto, Canada, kdenning@yorku.ca

Law Offices of Sterns and Tennen, Phoenix, Arizona, USA. ltennen@astrolaw.com

Sir Bernard Lovell Chair of Astrophysics, University of Manchester, Director of the Jodrell Bank Centre for Astrophysics, United Kingdom, michael.garrett@manchester.ac.uk

Abstract: In 1989, the International Academy of Astronautics (IAA) adopted the “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”, also known as the Search for Extraterrestrial Intelligence (SETI) Post-Detection Protocols. These provided a set of guiding principles to be followed in the event of a putative signal detection, having been developed by the IAA SETI Permanent Committee over a period of several years starting in the late 1980s, and subsequently approved by the International Institute of Space Law (IISL), and endorsed by the Committee on Space Research (COSPAR), the International Astronomical Union (IAU), and the (International Astronautical Federation (IAF). A streamlined update to reflect the emergence of the Internet was developed and accepted by the IAA SETI Permanent Committee in 2010 as the “Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence”.

Since then, the nature of global communication has been dramatically changed by social media, and multiple SETI ‘false alarms’ have demonstrated the ways in which information moves in our present era. Simultaneously, SETI searches – now also known techno-signature searches – have radically expanded in extent, scope, and power, with substantial search efforts underway at Breakthrough Listen, the SETI Institute, the Chinese Five-hundred-meter Aperture Spherical Telescope (FAST), and many other observatories around the world, between them using many of the world’s most powerful telescopes. Moreover, upcoming next-generation telescopes will allow the detection of molecules indicative of the existence of life in exoplanet atmospheres, conduct high cadence all-sky surveys and be able to detect the equivalent of an airport radar 100 light years away. As a result, the community of scientists engaged in SETI research is expanding significantly. In addition, other methods of detecting exoplanets as potential techno-signature targets are increasing, broadening the community of scientists who may not regard themselves as SETI-related but could potentially uncover data of SETI interest.

In 2022, the IAA SETI Committee decided that a high priority was to update the 2010 IAA Declaration of Principles. A proposal for a small task group with the sole task of updating these Principles was initiated at the end of 2022, and co-chairs put in place to develop that idea. Given rapid changes in SETI/technosignatures research and the dynamic communications landscape, we now envision an eventual combination of timeless Principles which generally preserve the original intentions of the 1989 Protocols and Codes of Conduct and Best Practices documents to provide additional regularly updated guidelines, responsive to changing circumstances. We present a report on the progress on the formation of the task group and its deliberations.